



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Discover. Learn. Empower.

Experiment:1

Student Name: Nitin Upadhyay

Branch: B.E. CSE

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Subject Name: System Design

UID:23BCS12270

Section/Group:KRG-3A

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Subject Code:23CSH-314

1. Aim: To design, implement, and evaluate a simple URL redirection system that converts long URLs into short URLs, and to measure system performance (latency and throughput) under varying request loads.

2. Objective:

- To design and implement a URL shortening and redirection system that maps long URLs to short URLs accurately.
- To ensure reliable redirection from short URLs to their original long URLs with minimal delay.
- To evaluate system performance by measuring latency and throughput under varying request loads.

3. Tools: Tiny URL, Browser ,Microsoft Excel

4. Procedure:

- Design the URL Shortening Logic
- Accept long URLs as input and generate unique short codes.
- Store long and short URL mappings in memory or a database.
- Create an endpoint to handle short URL requests.
- Redirect users to the original URL using HTTP 301/302 responses.
- Record request start and end times to calculate latency.
- Measure throughput by counting requests served per second.
- Perform load testing with different levels of concurrent users.

5. Implementation:

Latency:

It is the time delay between sending a request and receiving a response.

It is measured in units of milliseconds (ms) or seconds (s).

Throughput :

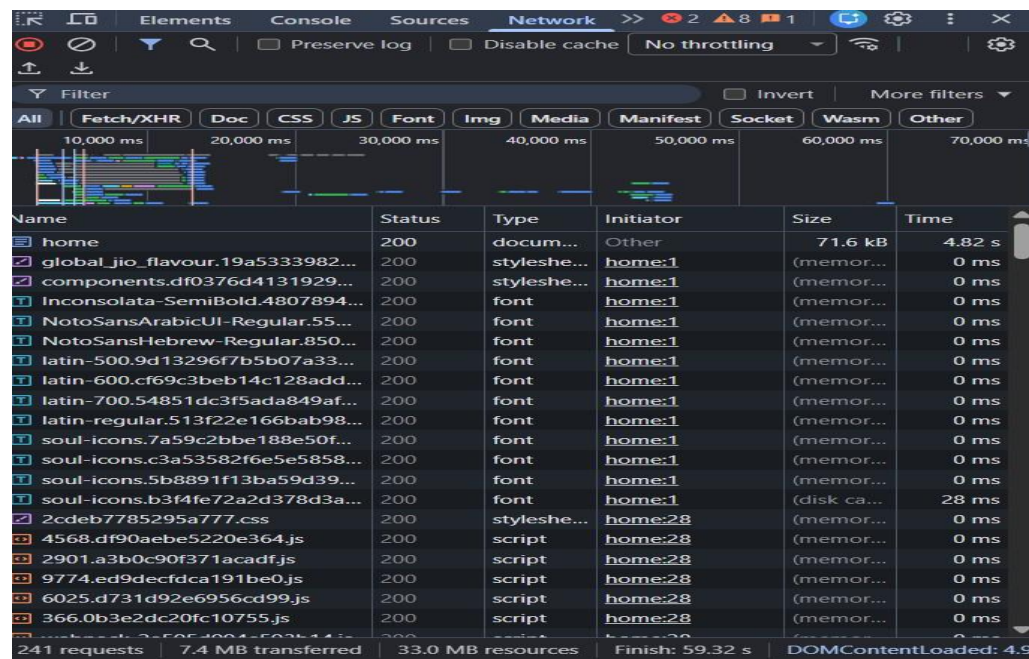
It is the number of requests or operations a system can successfully process in a given amount of time. It is measured in units of requests per second (RPS) or transactions per second (TPS).

Throughput = Total number of requests ÷ Total time taken

6. Output:

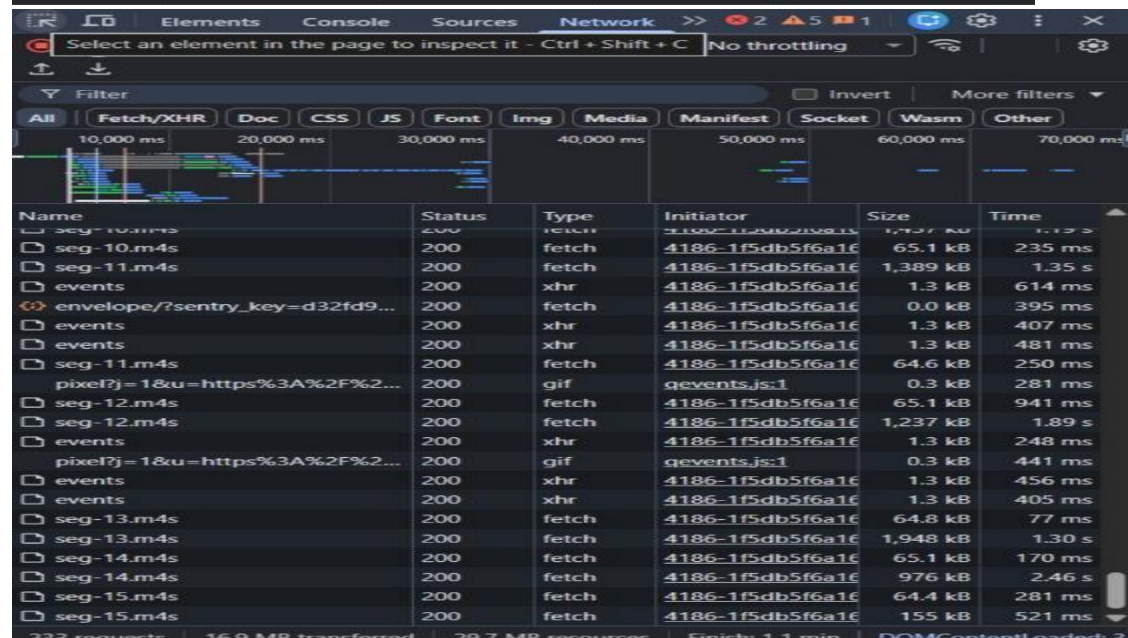
Website	Long-URL	Latency	Request	Finish Time	Throughput	Short-URL	Latency	Request	Finsh-Time	Throughput
Hotstar	https://www.hotstar.com/in/home	4.82	241	59.32	4.06	https://tinyurl.com/32u4yc37	302	233	1.1min	3.53
ww3school	https://www.w3schools.com/	2.27	53	7.27	7.29	https://tinyurl.com/5b339sck	601	53	5.58	9.45
MDN	https://developer.mozilla.org/en-US/	343	76	6.08	12.5	https://tinyurl.com/5dfx3m6r	387	76	5.69	13.3
Microsoft	https://www.microsoft.com/en-in	210	169	15.77	10.71	https://tinyurl.com/4p6jmta3	305	179	32.77	5.46
Netflix	https://www.netflix.com/	355	61	7.4	8.24	https://tinyurl.com/vyvezbuss	348	61	11.89	5.13

Hotstar:



Name	Status	Type	Initiator	Size	Time
home	200	docum...	Other	71.6 kB	4.82 s
global_jio_flavour.19a5333982...	200	styleshe...	home:1	(memor...	0 ms
components.df0376d4131929...	200	styleshe...	home:1	(memor...	0 ms
Inconsolata-SemiBold.4807894...	200	font	home:1	(memor...	0 ms
NotoSansArabicUI-Regular.55...	200	font	home:1	(memor...	0 ms
NotoSansHebrew-Regular.850...	200	font	home:1	(memor...	0 ms
latin-500.9d13296f7b5b07a33...	200	font	home:1	(memor...	0 ms
latin-600.cf69c3beb14c128add...	200	font	home:1	(memor...	0 ms
latin-700.54851dc3f5ada849af...	200	font	home:1	(memor...	0 ms
latin-regular.513f22e166bab98...	200	font	home:1	(memor...	0 ms
soul-icons.7a59c2bbe188e50f...	200	font	home:1	(memor...	0 ms
soul-icons.c3a53582f6e5e5858...	200	font	home:1	(memor...	0 ms
soul-icons.5b8891f13ba59d39...	200	font	home:1	(memor...	0 ms
soul-icons.b3f4fe72a2d378d3a...	200	font	home:1	(disk ca...	28 ms
2cdeb7785295a777.css	200	styleshe...	home:28	(memor...	0 ms
4568.df90aeb5220e364.js	200	script	home:28	(memor...	0 ms
2901.a3b0c90f371acadf.js	200	script	home:28	(memor...	0 ms
9774.ed9decfdca191be0.js	200	script	home:28	(memor...	0 ms
6025.d731d92e6956cd99.js	200	script	home:28	(memor...	0 ms
366.0b3e2dc20fc10755.js	200	script	home:28	(memor...	0 ms

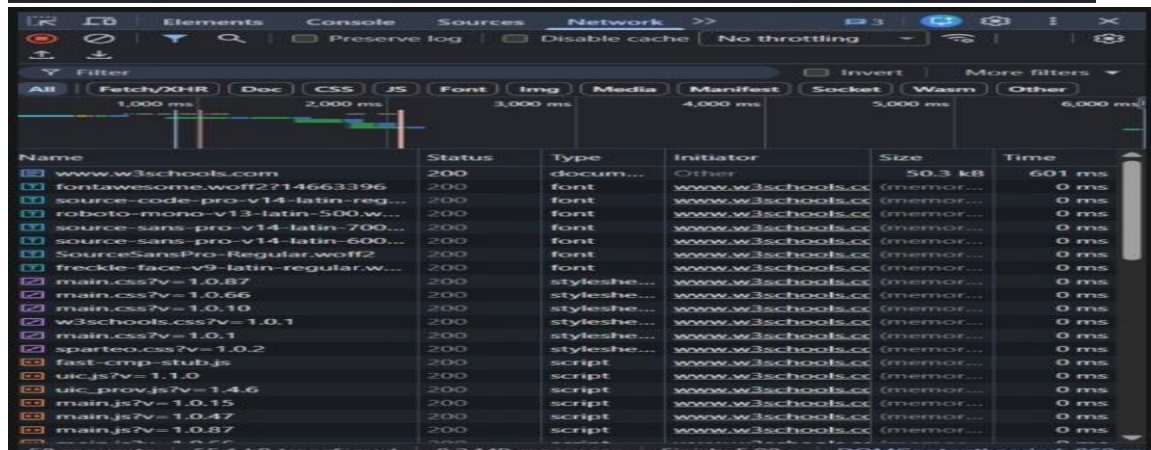
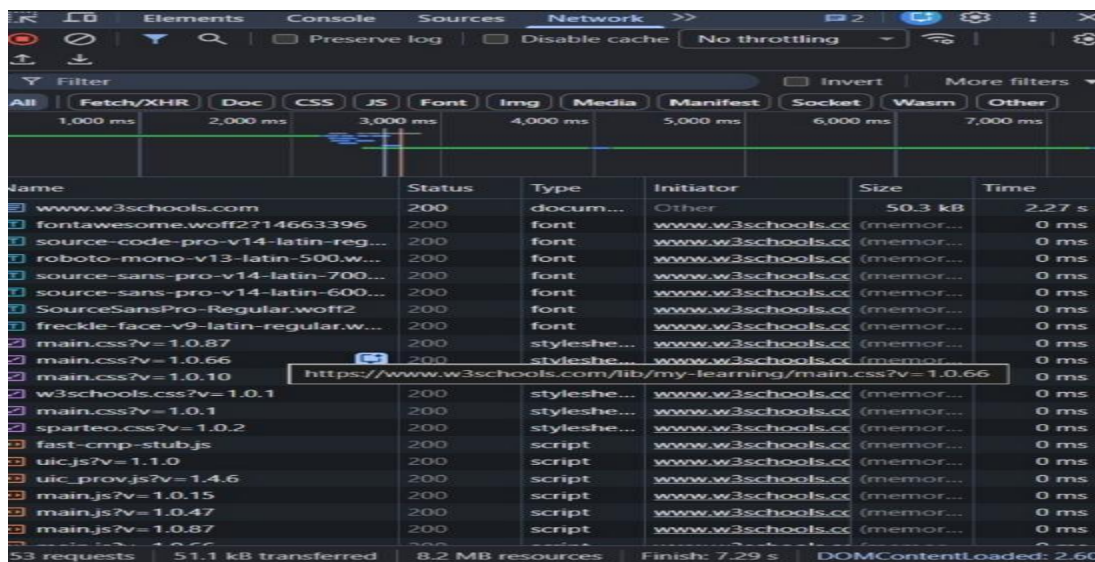
241 requests | 7.4 MB transferred | 33.0 MB resources | Finish: 59.32 s | DOMContentLoaded: 4.9



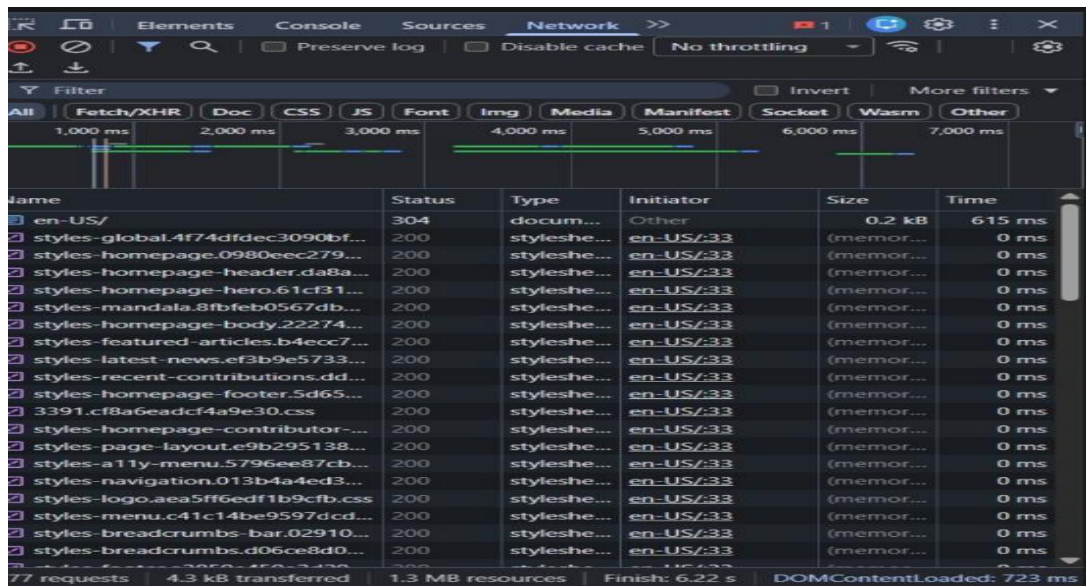
Name	Status	Type	Initiator	Size	Time
seg-10.m4s	200	fetch	4186-1f5db5f6a16	65.1 kB	235 ms
seg-11.m4s	200	fetch	4186-1f5db5f6a16	1,389 kB	1.35 s
events	200	xhr	4186-1f5db5f6a16	1.3 kB	614 ms
envelope/?sentry_key=d332fd9...	200	fetch	4186-1f5db5f6a16	0.0 kB	395 ms
events	200	xhr	4186-1f5db5f6a16	1.3 kB	407 ms
events	200	xhr	4186-1f5db5f6a16	1.3 kB	481 ms
seg-11.m4s	200	fetch	4186-1f5db5f6a16	64.6 kB	250 ms
pixel?j=1&u=https%3A%2F%2...	200	gif	qevents.js:1	0.3 kB	281 ms
seg-12.m4s	200	fetch	4186-1f5db5f6a16	65.1 kB	941 ms
seg-12.m4s	200	fetch	4186-1f5db5f6a16	1,237 kB	1.89 s
events	200	xhr	4186-1f5db5f6a16	1.3 kB	248 ms
pixel?j=1&u=https%3A%2F%2...	200	gif	qevents.js:1	0.3 kB	441 ms
events	200	xhr	4186-1f5db5f6a16	1.3 kB	456 ms
events	200	xhr	4186-1f5db5f6a16	1.3 kB	405 ms
seg-13.m4s	200	fetch	4186-1f5db5f6a16	64.8 kB	77 ms
seg-13.m4s	200	fetch	4186-1f5db5f6a16	1,948 kB	1.30 s
seg-14.m4s	200	fetch	4186-1f5db5f6a16	65.1 kB	170 ms
seg-14.m4s	200	fetch	4186-1f5db5f6a16	976 kB	2.46 s
seg-15.m4s	200	fetch	4186-1f5db5f6a16	64.4 kB	281 ms
seg-15.m4s	200	fetch	4186-1f5db5f6a16	155 kB	521 ms

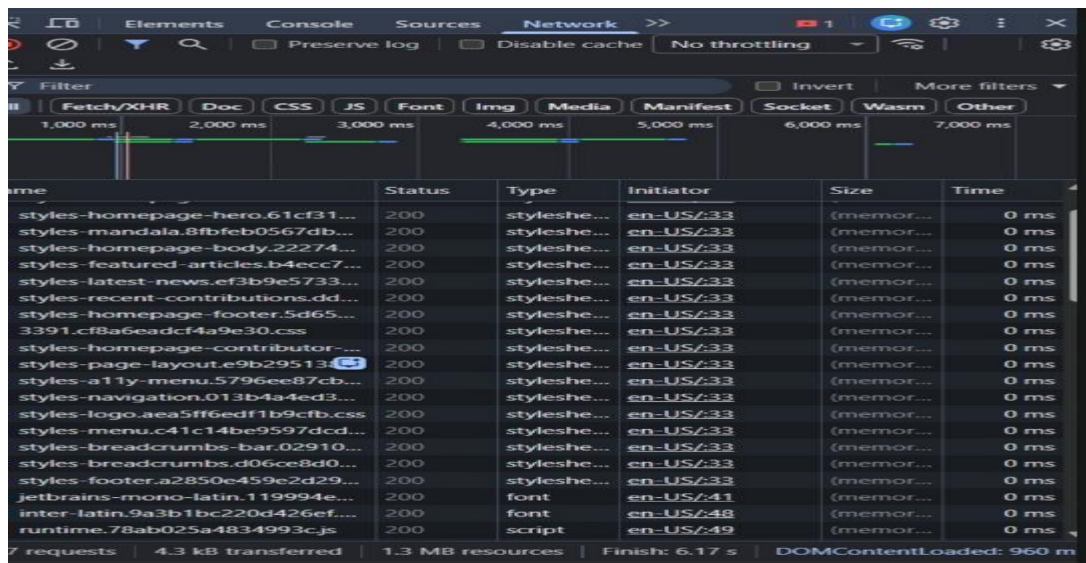
233 requests | 16.9 MB transferred | 29.7 MB resources | Finish: 1.1 min | DOMContentLoaded: 3

WW3school:

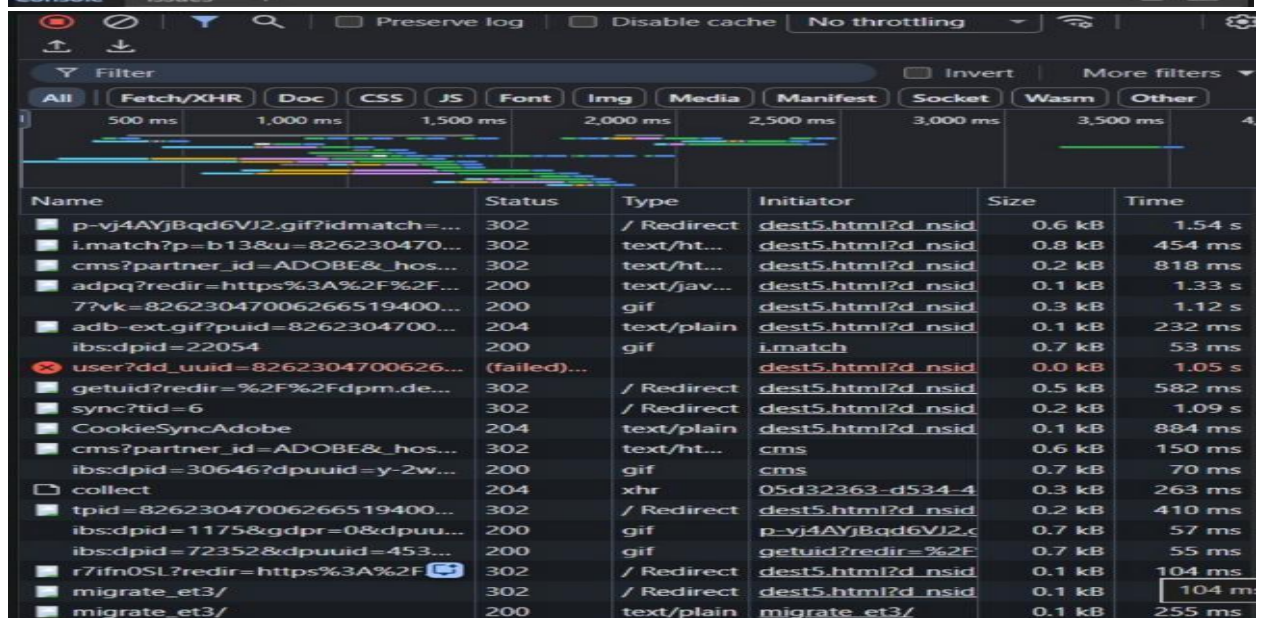
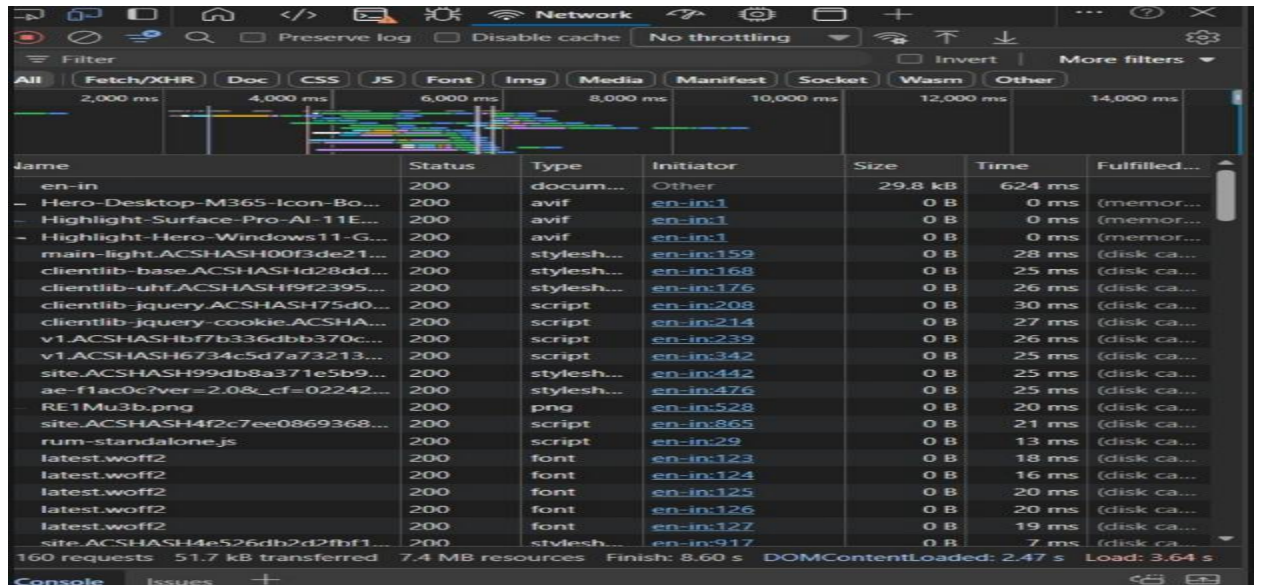


MDN:





Microsoft:



The figure consists of two screenshots of the Chrome DevTools Network tab, illustrating the process of analyzing web page loading performance.

Top Screenshot:

- Waterfall View:** A timeline showing the sequence of requests. The x-axis represents time in milliseconds, with markers at 5,000 ms, 10,000 ms, and 15,000 ms. The waterfall shows a series of requests, with a significant cluster of requests occurring between 5,000 ms and 10,000 ms.
- Request List:** A table listing the requests. The columns are Name, Status, Type, Initiator, Size, and Time. The list includes requests for various resources such as `in/`, `nmhpFrameworkClient.081e43...`, `otSDKStub.js`, `nmhp.1afe2e52bff213979e1d.c...`, `netflix-sans-core-variable-full...`, `IN-en-20260105-TRIFECTA-per...`, `2?fetchType=js&eventType=W...`, `2?fetchType=js&eventType=W...`, `87b6a5c0-0104-4e96-a291-09...`, `NetflixBans_W_Rg.woff2`, `NetflixBans_W_Md.woff2`, `NetflixBans_W_Bd.woff2`, `NetflixBans_W_Blk.woff2`, `location`, `IN-en-20260105-TRIFECTA-p...`, `AAAABXMINT8IRIEH5nkoYUHK...`, `AAAABaz9GSNM5MVZSPLChs...`, `AAAABd-hr2-TdXUEamf_D6vpl...`, `AAAABTf3e14TtZ2Z3216Uaph...`, and `AAAABVeIHlYADJlly5NwDA2dg...`.

Bottom Screenshot:

- Waterfall View:** A timeline showing the sequence of requests. The x-axis represents time in milliseconds, with markers at 2,000 ms, 4,000 ms, 6,000 ms, 8,000 ms, 10,000 ms, and 12,000 ms. The waterfall shows a series of requests, with a significant cluster of requests occurring between 4,000 ms and 8,000 ms.
- Request List:** A table listing the requests. The columns are Name, Status, Type, Initiator, Size, and Time. The list includes requests for various resources such as `in/`, `nmhpFrameworkClient.081e43...`, `otSDKStub.js`, `nmhp.1afe2e52bff213979e1d.c...`, `netflix-sans-core-variable-full...`, `NetflixBans_W_Rg.woff2`, `NetflixBans_W_Md.woff2`, `NetflixBans_W_Bd.woff2`, `NetflixBans_W_Blk.woff2`, `2?fetchType=js&eventType=W...`, `2?fetchType=js&eventType=W...`, `87b6a5c0-0104-4e96-a291-09...`, `location`, `IN-en-20260105-TRIFECTA-per...`, `otBannerSdk.js`, `en.json`, `otFlat.json`, `otPcTab.json`, `otCommonStyles.css`, `AAAABXMINT8IRIEH5nkoYUHK...`, and `AAAABaz9GSNM5MVZSPLChs...`.

7. Learning Outcome:

- Learned how to design a system that converts long URLs into unique and usable short URLs.
- Understood how HTTP redirection (301/302) enables fast and reliable URL forwarding.
- Gained knowledge of measuring latency to analyze system response time.
- Learned to calculate throughput to evaluate system request-handling capacity.